

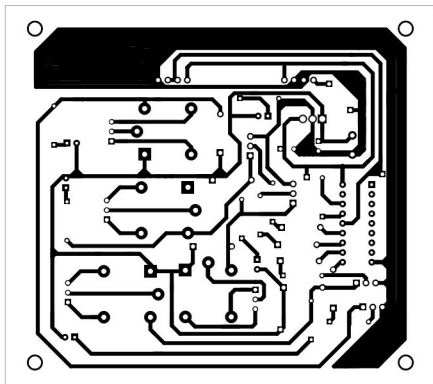


Fig. 9: Actual-size PCB layout of the receiver circuit

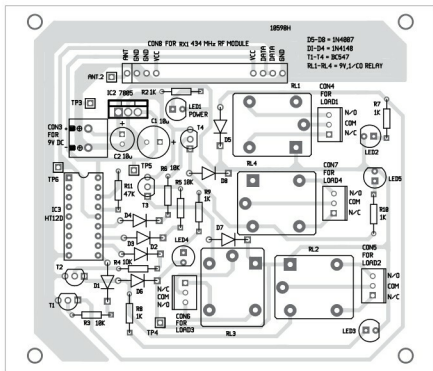


Fig. 10: Components layout for the PCB in Fig. 9

the project file, MATLAB will try to communicate with the board. After successful communication is estab-

lished, you can control the equipment by pressing the appropriate pushbutton in the GUI.

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download at source.efymag.com

Test Points

Test point	Expected value
TP0, TP3	0V
TP1	5V
TP2	Pulse train
TP4	9V
TP5	5V
TP6	Pulse train

Construction and testing

An actual-size PCB layout of the transmitter side is shown in Fig. 7 and its components layout in Fig. 8. Similarly, actual-size PCB layout of the receiver is shown in Fig. 9 and its components layout in Fig. 10.

Connect the transmitter unit to the PC. After uploading the Arduino code into Board1 as explained above, run the MATLAB GUI program. Next, connect 9V battery to the receiver unit. Keep transmitter and receiver unit one meter apart. From the GUI program, click 'on' or 'off' button to switch on/off the corresponding electrical device. The 'on' status of the devices is indicated by the glowing of LED2 through LED5.

For troubleshooting, the expected values/voltages at various test points are listed in the table above. **EFY**



Saikat Patra is passionate about electronics and MCU-based embedded system applications



Shibendu Mahata is an M.Tech (gold medalist) in instrumentation and electronics engineering from Jadavpur University. He has been interested in MCU-based real-time embedded signal processing and process control systems